

VisionGuard: Software Requirements Specification

Version: 1.0 (Prototype baseline) **Date:** 7 October 2026 **Prepared by:** [Team name / member names] **Status:** Draft for product development event **Standard followed:** IEEE 29148 / IEEE 830 structure, security controls mapped to OWASP MASVS v2

1. Introduction

1.1 Purpose

This document specifies the requirements for VisionGuard, an Android application that protects users' eyesight, screen privacy and time using on-device computer vision and local usage tracking. It is intended for developers, testers, evaluators and stakeholders.

1.2 Scope

VisionGuard is a privacy-first phone safety and wellbeing system. It combines existing technologies (on-device face detection, Android usage statistics, overlay windows) into one integrated, resource-efficient experience. It does not introduce new AI models. All processing is local; the prototype has no cloud component.

In scope: Eye Guard, Privacy Guard, Smart Dashboard, Multi-User Profiles, Time Tokens, local encrypted storage. **Out of scope (prototype):** cloud sync, remote parental control, face-based user identification, iOS support, content filtering.

1.3 Definitions and abbreviations

Term	Meaning
FGS	Foreground Service (Android)
Overlay	Window drawn above other apps via TYPE_APPLICATION_OVERLAY
Time Token	A short, limited usage window granted after a daily limit is reached
Shoulder surfing	A second person viewing the user's screen

Term	Meaning
EMA	Exponential moving average
MASVS / MASTG	OWASP Mobile Application Security Verification Standard / Testing Guide
DPDP	India's Digital Personal Data Protection Act, 2023

1.4 References

- IEEE 29148: Systems and software engineering, requirements engineering
- OWASP MASVS v2 and MASTG
- Android developer documentation: CameraX, ML Kit Face Detection, UsageStatsManager, foreground service types, overlay windows
- DPDP Act 2023 (India), COPPA (US), GDPR (EU), for consent considerations only

1.5 Document overview

Section 2 describes the product context. Section 3 lists specific requirements. Section 4 covers data. Section 5 covers security and privacy. Section 6 covers verification. Section 7 covers risks and prototype scope.

2. Overall description

2.1 Product perspective

VisionGuard is a standalone Android application. It relies on the device front camera, the Android usage-statistics service and the system overlay facility. It is independent of Google accounts and any server.

2.2 Product functions (summary)

ID	Function	Summary
F1	Eye Guard	Detects the phone held too close, warns, and dims the screen
F2	Privacy Guard	Detects an additional viewer, alerts, and dims or blurs the screen

ID	Function	Summary
F3	Multi-User Profiles	Tracks usage and limits per person (e.g., parent and child)
F4	Smart Dashboard	Shows usage and suggests alternatives to doom-scrolling
F5	Time Tokens	Replaces hard blocks with cooldowns and 5-minute windows

2.3 User classes

User	Description	Privileges
Adult user	Uses own device or a parent profile	Full settings, can set limits, view all profiles
Parent (administrator)	Manages child profiles	PIN/biometric-protected settings, limit and token configuration
Child user	Uses a restricted profile	Cannot change limits or disable protection
Evaluator / tester	Verifies privacy and security claims	Read-only inspection through the Privacy Ledger

2.4 Operating environment

- Android 10 (API 29) or higher; tested target: Android 13 and 14
- Device with front camera, minimum 3 GB RAM (mid-range class)
- Permissions granted by the user: Camera, Display over other apps, Usage access, Notifications
- Sideloaded APK for the prototype

2.5 Design and implementation constraints

- C1: Kotlin, Jetpack Compose, MVVM, Hilt, Coroutines/Flow.
- C2: The camera cannot be accessed from a plain background service. A camera-type foreground service with a persistent notification is mandatory (Android 14 requires the `FOREGROUND_SERVICE_CAMERA` permission).

- C3: Android 12+ blocks touch pass-through for overlays above 0.8 opacity, so the dim effect is capped at 0.8.
- C4: Blur-behind is device-dependent; a frosted/dark overlay is the fallback.
- C5: Android's native multi-user is not available to normal apps, so profiles are app-level.
- C6: Some OEM skins (Xiaomi, Oppo, Vivo, Samsung) may terminate background services.
- C7: `AccessibilityService` shall not be used (Play Store restrictions and larger attack surface).

2.6 Assumptions and dependencies

- A1: Users grant required permissions.
- A2: ML Kit Face Detection (bundled model) is available offline.
- A3: Distance estimates are approximate (target error within about 15 percent after calibration).
- A4: A parent sets up child profiles; the child does not have the parent PIN.

3. Specific requirements

Priority key: **M** = must (prototype), **S** = should, **C** = could (roadmap). Requirement language: "shall" is mandatory.

3.1 Functional requirements

3.1.1 Eye Guard (F1)

ID	Requirement	Priority
FR-EG-1	The system shall capture front-camera frames at low resolution (320x240 to 640x480) using CameraX <code>ImageAnalysis</code> with the keep-only-latest strategy.	M
FR-EG-2	The system shall detect the largest face per frame using an on-device model and compute its pixel width.	M

ID	Requirement	Priority
FR-EG-3	The system shall estimate viewing distance using the pinhole model, calibrated by a one-time user action ("hold phone at 30 cm and tap Calibrate").	M
FR-EG-4	The system shall smooth estimates with an EMA and trigger only after N consecutive frames below the threshold (default N = 5) to avoid flicker.	M
FR-EG-5	The default "too close" threshold shall be 30 cm, user-adjustable, with 5 cm hysteresis before the warning clears.	M
FR-EG-6	On trigger, the system shall display a warning notification or banner and apply a dim overlay proportional to closeness, capped at 0.8 opacity.	M
FR-EG-7	The dim shall be removed within 1 second after the distance returns above threshold plus hysteresis.	M
FR-EG-8	The system shall stop analysing when the screen is off or the device is locked.	M
FR-EG-9	The system should read camera focal length and sensor size from <code>CameraCharacteristics</code> for automatic calibration.	S
FR-EG-10	The system could use iris or inter-pupillary landmarks for improved accuracy.	C

3.1.2 Privacy Guard (F2)

ID	Requirement	Priority
FR-PG-1	The system shall detect multiple faces in a frame.	M
FR-PG-2	A secondary viewer shall be recognised when a second face exceeds a minimum relative size (default 10 percent of frame width, tunable) and its head yaw indicates it faces the screen.	M
FR-PG-3	The condition shall persist for N consecutive frames (default 3) before alerting.	M

ID	Requirement	Priority
FR-PG-4	On trigger, the system shall alert the user and apply a dark or frosted overlay (blur-behind where supported).	M
FR-PG-5	The overlay shall clear automatically when the second face is absent for 2 seconds, or when the user taps a dismiss control.	M
FR-PG-6	The user shall be able to enable or disable Privacy Guard per profile.	M
FR-PG-7	The system could support a "trusted viewer" mode that suppresses alerts temporarily.	C

3.1.3 Overlay and display control

ID	Requirement	Priority
FR-OV-1	The system shall draw overlays using <code>TYPE_APPLICATION_OVERLAY</code> with <code>FLAG_NOT_TOUCHABLE</code> so underlying apps remain usable.	M
FR-OV-2	The system shall request the overlay permission with a clear explanation and a direct link to the settings screen.	M
FR-OV-3	The system shall show a persistent foreground notification while monitoring is active, with a one-tap pause action.	M
FR-OV-4	The system could adjust real screen brightness via <code>WRITE_SETTINGS</code> as an alternative to overlay dimming.	C

3.1.4 Multi-User Profiles (F3)

ID	Requirement	Priority
FR-MU-1	The system shall support at least two local profiles (e.g., Parent, Child), each with separate usage counters, limits and settings.	M
FR-MU-2	Profile switching shall require a PIN; parent-only actions shall require PIN or biometric authentication.	M
FR-MU-3	Usage shall be attributed to the active profile at the time it occurs.	M
FR-MU-4	Child profiles shall not be able to change limits, disable protection or view other profiles.	M
FR-MU-5	The system could identify the active user by face. This requires explicit opt-in, an encrypted embedding store and no export.	C

3.1.5 Smart Dashboard (F4)

ID	Requirement	Priority
FR-DB-1	The system shall read per-app foreground time using <code>UsageStatsManager (queryEvents)</code> after the user grants usage access.	M
FR-DB-2	The dashboard shall show today's total screen time, per-app breakdown, and counts of Eye Guard and Privacy Guard events, for the active profile.	M
FR-DB-3	The system shall generate rule-based suggestions (e.g., after 40 continuous minutes in a feed app, suggest a walk, stretch or reading).	M
FR-DB-4	The dashboard should show a 7-day trend.	S

3.1.6 Time Tokens (F5)

ID	Requirement	Priority
FR-TT-1	The system shall implement the state machine <code>ACTIVE -> LIMIT_REACHED -> COOLDOWN -> TOKEN_WINDOW (5 min) -> LIMIT_REACHED .</code>	M
FR-TT-2	When a per-app or per-category limit is reached, the system shall show a block screen rather than permanently disabling the app.	M
FR-TT-3	After a configurable cooldown (default 15 minutes), the user shall be able to redeem one Time Token for a 5-minute window.	M
FR-TT-4	The maximum tokens per day per profile shall be configurable by the parent (default 3).	M
FR-TT-5	Timing logic shall use <code>SystemClock.elapsedRealtime()</code> so changing the system clock does not bypass limits, and state shall persist across reboots.	M
FR-TT-6	Limit and token logic shall reside in a pure-Kotlin PolicyEngine with no Android dependencies.	M
FR-TT-7	The block screen shall be shown via the overlay permission, because starting activities from the background is restricted.	M

3.2 External interface requirements

Interface	Requirement
User interface	Jetpack Compose, Material 3, light and dark themes, minimum touch target 48 dp, readable at 1.0 and 1.3 font scale
Hardware	Front camera; optional accelerometer, light and proximity sensors for power gating

Interface	Requirement
Software	Android framework APIs listed in 2.5; ML Kit Face Detection (bundled) or MediaPipe Face Landmarker
Communications	None. The prototype shall not declare the <code>INTERNET</code> permission.

3.3 Non-functional requirements

Performance and efficiency

- NFR-P1: Per-frame detection latency shall be 100 ms or less on a mid-range device.
- NFR-P2: Eye Guard response (from crossing threshold to dim shown) shall be 1 second or less.
- NFR-P3: Adaptive frame rate: about 1 fps when idle, up to 5 fps near thresholds or when multiple faces are suspected.
- NFR-P4: Monitoring shall run only while the screen is on and unlocked.
- NFR-P5: Added battery drain should be measured and reported (target under 5 percent per hour of active monitoring; to be validated, not yet measured).
- NFR-P6: App memory footprint should stay under 150 MB.

Reliability

- NFR-R1: The service shall restart cleanly after crash, process death or reboot (with user consent for autostart).
- NFR-R2: If the camera is unavailable or in use by another app, the system shall degrade gracefully and inform the user.
- NFR-R3: The app shall guide users through OEM autostart and battery-optimisation exemptions.

Usability

- NFR-U1: First-run setup (permissions plus calibration) shall take under 3 minutes.
- NFR-U2: Warnings shall be non-punitive and clearly explain why the screen dimmed.
- NFR-U3: The camera-active indicator and foreground notification shall never be hidden or suppressed.

Compatibility and maintainability

- NFR-C1: Supported on Android 10 or higher; verified on at least two OEM skins.
- NFR-M1: Modular architecture (`core-vision` , `core-overlay` , `core-usage` , `core-policy` , `core-data` , `feature-dashboard` , `feature-profiles`).
- NFR-M2: PolicyEngine unit-test coverage of 90 percent or more of branches.

4. Data requirements

Data	Stored?	Location	Protection
Camera frames / bitmaps	Never	Memory only, released immediately (<code>ImageProxy.close()</code>)	Not logged, not written to disk
Face embeddings	Not in prototype	n/a	Roadmap: encrypted, opt-in, non-exportable
Derived events (e.g., "too close, 14:03")	Yes	Local database	Encrypted at rest
Usage statistics	Derived summaries	Local database	Encrypted at rest
Profiles, limits, token state	Yes	Local database	Encrypted at rest; PIN stored as salted hash
Calibration value	Yes	Local database	Encrypted at rest

Retention shall default to 30 days for events, configurable, with a one-tap "Wipe all data" action.

5. Security and privacy requirements

5.1 Privacy by architecture

- SEC-P1: The manifest shall not declare `INTERNET` or `ACCESS_NETWORK_STATE` ; "no cloud" is therefore verifiable, not a promise.

- SEC-P2: Frames shall be processed in memory only and never persisted, cached, logged or shared.
- SEC-P3: Only derived events shall be stored.
- SEC-P4: The app shall provide a "Privacy Ledger" screen listing exactly what is stored, plus a wipe control.
- SEC-P5: `allowBackup` shall be false and data-extraction rules shall exclude all app data.
- SEC-P6: Explicit consent shall be shown before enabling camera analysis, with parental consent flow for child profiles.

5.2 Data protection and authentication

- SEC-D1: Database encryption with keys held in Android Keystore (StrongBox where available), using Room with SQLCipher or Tink. Deprecated `androidx.security:security-crypto` shall not be used.
- SEC-D2: PINs shall be hashed with PBKDF2 or Argon2 using a unique salt, with attempt rate-limiting and lockout back-off.
- SEC-D3: Parent actions shall support `BiometricPrompt` with a `CryptoObject`.
- SEC-D4: `FLAG_SECURE` shall be set on dashboard and settings screens to prevent screenshots and screen recording.

5.3 Application hardening

- SEC-H1: All components shall be `exported="false"` unless required.
- SEC-H2: `usesCleartextTraffic="false"` and a restrictive network security config shall be set for future-proofing.
- SEC-H3: R8 minification enabled and release builds properly signed; debug flags off.
- SEC-H4: Dependencies shall be scanned (Dependabot or OWASP Dependency-Check) and an SBOM generated.
- SEC-H5: Settings shall be PIN-protected; Device Admin uninstall protection is a roadmap item.
- SEC-H6: Overlay and block screens shall protect against tapjacking by not relying on touch events from untrusted windows for sensitive actions.

5.4 Threat model summary (STRIDE)

Threat	Example	Mitigation
Spoofing	Child guesses the parent PIN	Rate limiting, lockout, biometric option
Tampering	Child changes system clock to bypass limits	Monotonic clock, persisted state
Repudiation	Disputed usage records	Append-only local event log
Information disclosure	Extracting database or frames from the device	Encryption, no frame persistence, no backup, <code>FLAG_SECURE</code>
Denial of service	Android or OEM kills the service	FGS, exemption guidance, restart logic
Elevation of privilege	Malicious app invokes exported components	No exported components, signature-level permissions

5.5 Legal and compliance note

Processing facial imagery and monitoring minors' usage may trigger consent obligations under the DPDP Act 2023 (India), COPPA and GDPR. Local-only processing reduces exposure but does not remove these obligations. Qualified legal review is required before any public release. This SRS is not legal advice.

6. Verification and acceptance

Requirement area	Method	Pass criterion
SEC-P1 (no network)	Manifest inspection; PCAPdroid or Wireshark capture via hotspot	<code>INTERNET</code> absent; zero outbound packets during a 10-minute run
SEC-P2 (no frames stored)	File-system inspection after run; code review	No image files; no frame data in logs
SEC-D1 (encryption)	Pull DB from device; Frida/objection inspection of own build	Database unreadable without key

Requirement area	Method	Pass criterion
SEC-H1 (exports)	MobSF / drozer scan	No unintended exported components
FR-EG-3 to 7	Manual test at 20, 30, 40 and 60 cm	Warning and dim within 1 s at under 30 cm; clears when back
FR-PG-2 to 5	Manual test with a second person behind the user	Alert after 3 frames; clears after 2 s absence
FR-TT-5	Change system clock; reboot	Limits not bypassed; state preserved
FR-TT-1 to 4	Unit tests on PolicyEngine	All state transitions covered
NFR-P5	Perfetto / Battery Historian over 1 hour	Drain figure recorded
NFR-C1	Test on two OEM devices	Service survives 30 minutes in background

Standards checklist: OWASP MASVS v2 (L1 baseline plus L2 for storage and authentication), tested per MASTG.

7. Risks, prototype scope and roadmap

7.1 Key risks

Risk	Impact	Mitigation
Camera or overlay fails in background on some devices	Core demo fails	Day-one spike on a real device; backup screen recording
Distance estimate inaccurate	False warnings	Per-user calibration, smoothing, hysteresis
Battery drain from camera	Poor adoption	Adaptive fps, screen-on gating, sensor gating
OEM background kills	Protection silently stops	Autostart guidance, service health indicator

Risk	Impact	Mitigation
Child bypasses or uninstalls app	Limits ineffective	PIN-protected settings; Device Admin (roadmap)
Privacy or consent non-compliance	Legal exposure	Local-only design, consent flows, legal review

7.2 Prototype scope (event build)

Feature	Status in prototype
Eye Guard	Implemented (calibrated distance, dim overlay, warning)
Privacy Guard	Implemented (multi-face detection, overlay)
Smart Dashboard	Implemented with real usage data and rule-based suggestions
Multi-User Profiles	Partial (PIN switch, two profiles, separate counters)
Time Tokens	Mocked UI of the state machine; PolicyEngine as time permits
Encryption, Privacy Ledger, hardening	Partial; remainder on roadmap

7.3 Roadmap

1. Full PolicyEngine and persistence for Time Tokens
2. Keystore-backed database encryption and complete MASVS checklist
3. Automatic calibration and iris-based distance
4. Trusted-viewer mode for Privacy Guard
5. Device Admin uninstall protection and OEM battery handling
6. Optional local-first parent-to-child sync (Nearby Connections or mDNS with mutual-auth TLS or Noise, end-to-end encrypted, no central server)
7. Play Store declarations (camera FGS, overlay, usage access) and legal review

Appendix A: Time Token state machine

From	Event	To
ACTIVE	Daily limit reached	LIMIT_REACHED
LIMIT_REACHED	Block screen shown, cooldown starts	COOLDOWN
COOLDOWN	Cooldown elapsed and token available	TOKEN_WINDOW (5 min)
TOKEN_WINDOW	5 minutes elapsed	LIMIT_REACHED
LIMIT_REACHED / COOLDOWN	Daily reset or parent override	ACTIVE

Appendix B: Open items

- Final threshold and N-frame values to be tuned on the test device.
- Calibration method and accuracy to be validated across at least 3 users.
- Battery benchmark results to be added after testing.
- Team names, roles and sign-off to be completed in the group format.